

How America Talks

Christine Lee, Chris Yang, Nathanael Choi, & Julie Hohenberg
Advisor: Professor Prud'Hommeaux



Motivation

- Americans disagree on everything—even whether to say "sub," "hoagie," or "grinder"
- We analyze dialect trends and predict regions of origin.

Data

- Decades-long dialect survey (Dr. Bert Vaux) with respondent demographics
- Removed duplicates + non-continental respondents; assigned region labels
- 108 questions, 47K+ respondents, 43-47K answers/question

Survey Question

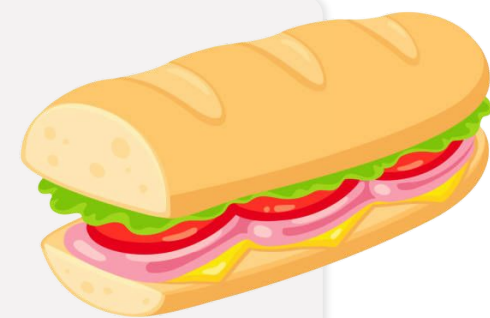
What do you call the long cold sandwich that contains cold cuts, lettuce, and so on?

Answer Choices

Grinder, hoagie, hero, sub (submarine) ...

Respondent Answer

Sub (submarine)



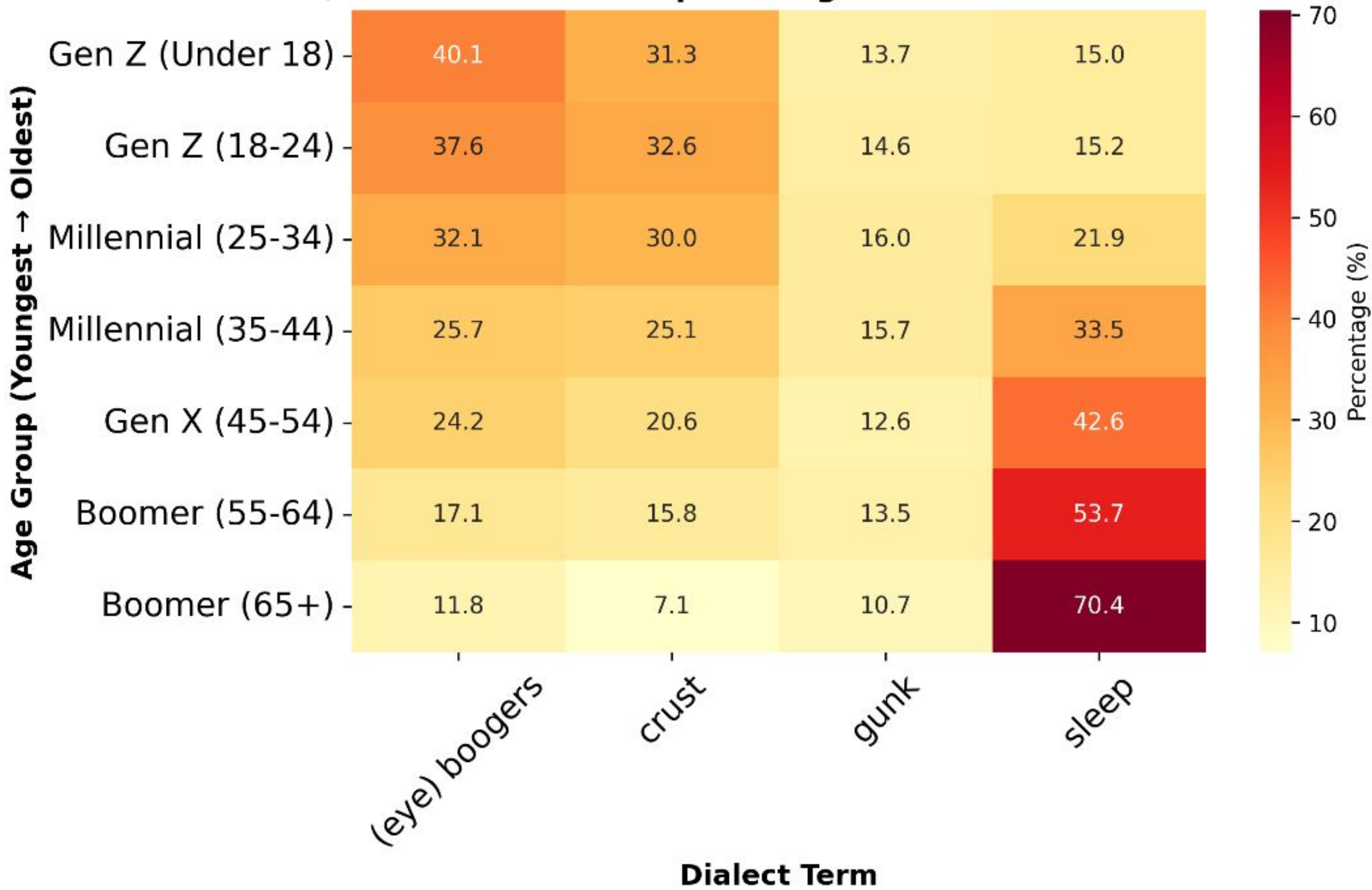
Analysis: Change in Dialect Through Time

Q: What do you call gooey or dry matter that collects in the corners of your eyes?

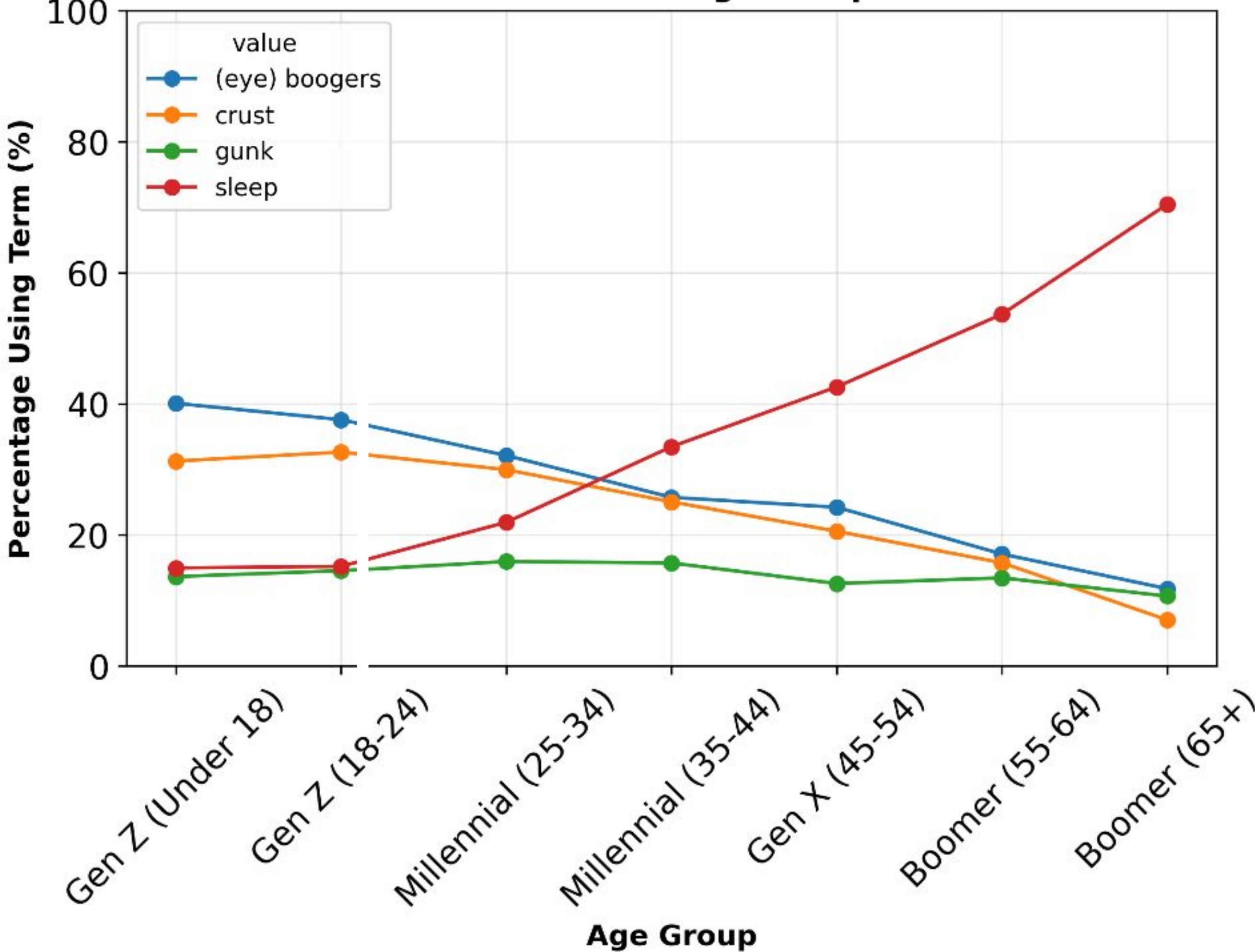
$X^2 = 134,222.8$, $p=0.0000$, Cramer's V = 0.369

- a) X^2 (large): pattern exists
b) P-value (low): confidence
c) Cramér's V: moderate strength

Heatmap of Usage Patterns



Trend Across Age Groups



Region Prediction Quiz

Method

- Dropped lat/long from training data
- 80/20 test-train split
- Target variable: dialect region labels

Results

Model (Sklearn)	Accuracy
Logistic Regression	0.57
Random Forest	0.53
Random Baseline	0.17

Quiz showcased in Dashboard (Live Demo 12/8)

Dashboard (Live Demo 12/8)

Tech Stack

- Host data in Google Drive
- Version control with GitHub
- Develop in VS Code
- Deploy on Streamlit Cloud

Content

- Interactive visualizations
- Try a dialect prediction quiz
- Learn about English dialects in North America

Conclusion & Next Steps



- Generational shifts in usage

- Language can predict region



- Multi-region predictions

- Improve + release dashboard



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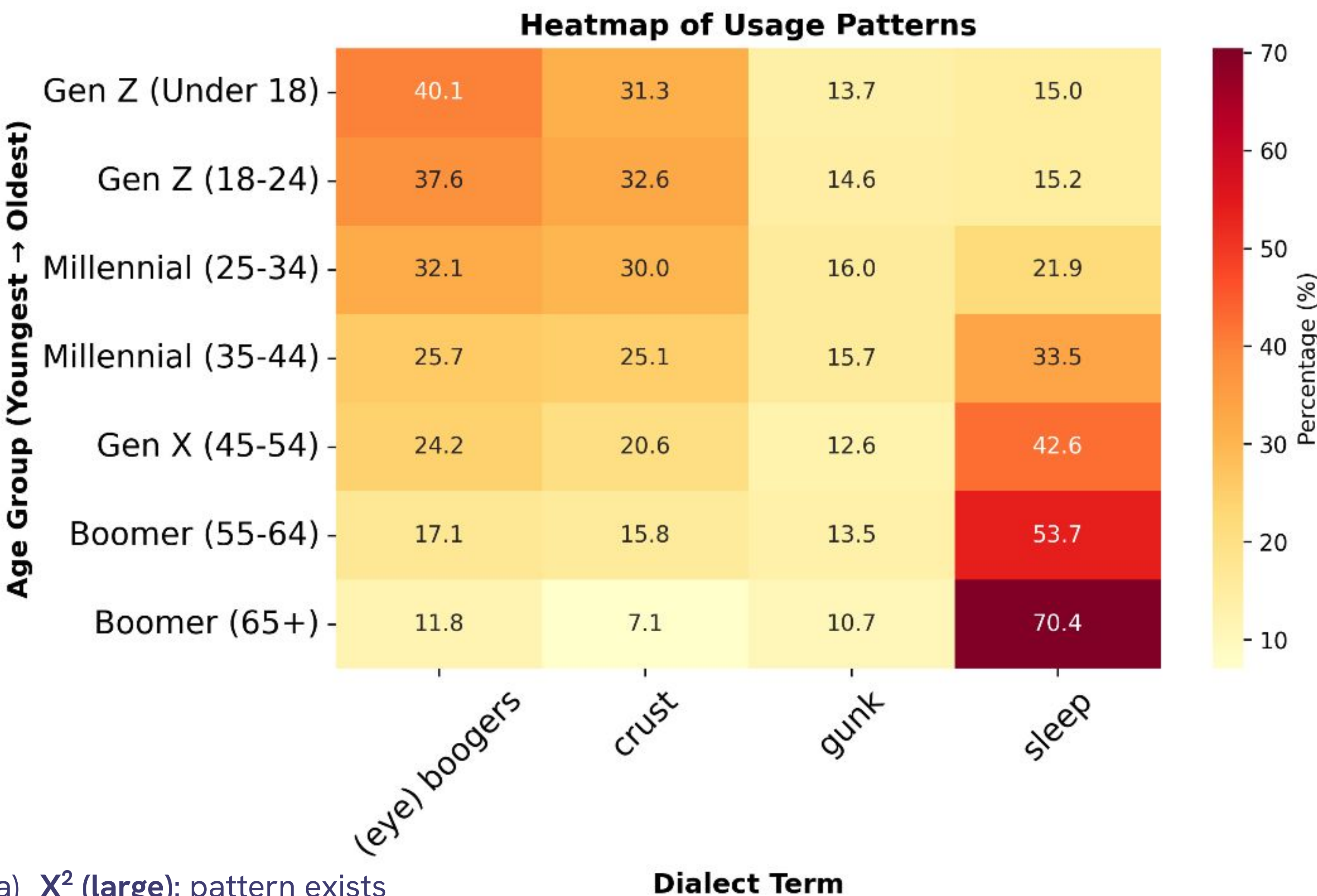
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Motivation

Soda!

Pop!

Region Prediction Quiz (Dashboard)

Dashboard (Live Demo 12/8)

- **Americans disagree on everything**—even whether to say "sub," "hoagie," or "grinder"...
- Using these linguistic quirks, we analyze dialect trends on a **dashboard** and build an **interactive quiz** that predicts a speaker's likely origin based on word choice!

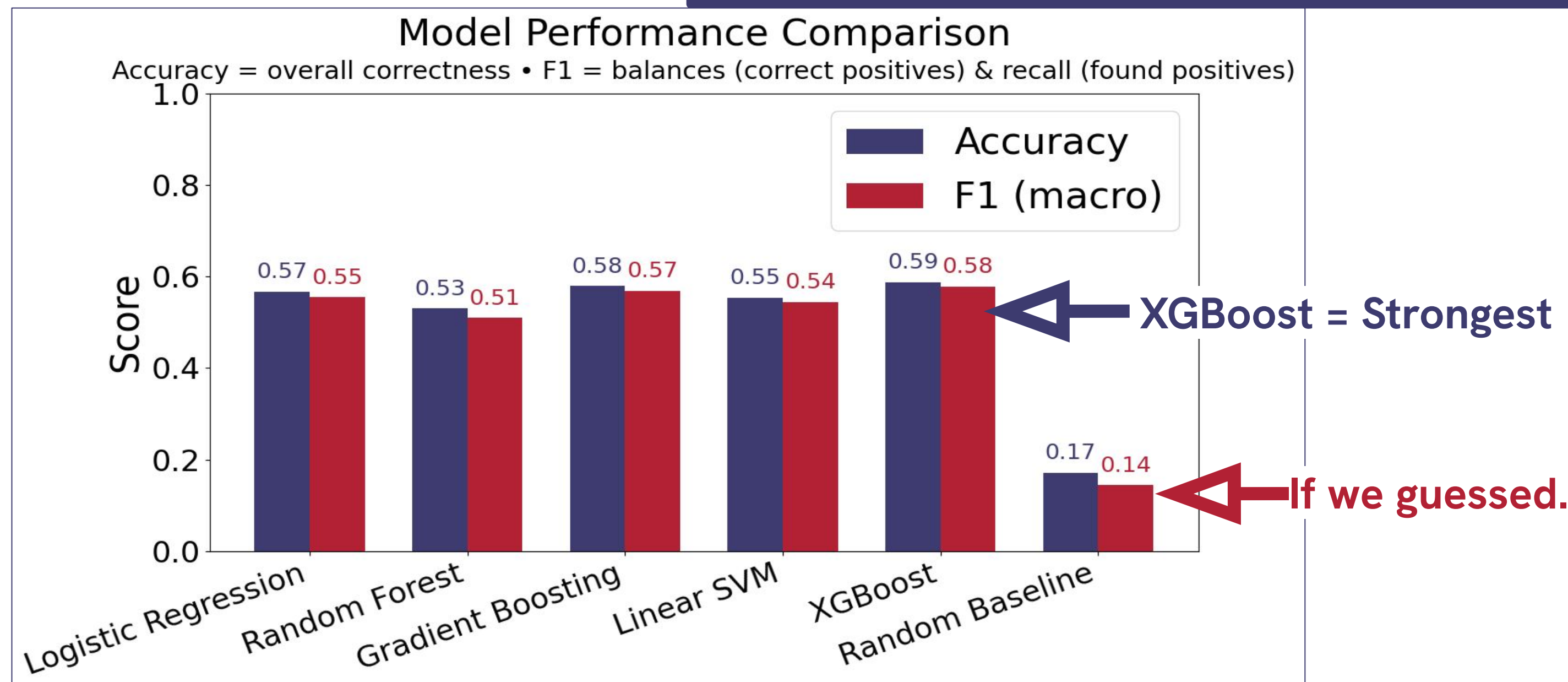
Data

- Data collected by **Dr. Bert Vaux** through **decades of English dialect surveys**
- **108 questions** (i.e., "Do you say *"expecially"*, or *"especially"*?")
- **47K+** respondents, **43–47K** responses/question
- Includes **respondent demographics** (i.e., age, location, native language)
- We **removed duplicates** and kept respondents in **continental United States**
- Assigned ground-truth **region labels** by lat/long:

Data Processing

Goal: Predict a user's region of origin based on their word choice

- One-hot-encoded **question/answer pairs**
- **80/20** test-train split
- Target variable: **dialect region labels**



Region	Strongest Predictor
Greater New York City	calling a milk-and-ice-cream drink a “cabinet”
Midland	referring to DC as “the City”
North Central	calling the end of a loaf of bread the “nose”
Northern New England	calling the night before Halloween “cabbage night”
The North	calling the night before Halloween “goosy night”
The South	calling the highway median “neutral ground”
The West	calling the long-legged spider “father longlegs”

Tech Stack

- Host data in **Google Drive**
- Version control with **GitHub**
- Develop in **VS Code**
- Deploy on **Streamlit Cloud**
- **Packages** used: Streamlit, Pandas, numpy, plotly, gdown, re, pathlib

Content

- **Interactive** visualizations
- **Try** a dialect prediction quiz
- **Learn** about English dialects in North America

English Dialect App

Explore our detailed datasets of dialect variation in the English-speaking world.

Data Source

Dr. Bert Vaux, a linguistics professor from the University of Cambridge, collected English dialect data for decades.

- Early surveys were done on **paper** and manually transcribed
- Later surveys were collected **digitally** by CS students
- Questions include both **text prompts** and **images**

This dataset reflects natural, real-world variation, but also comes with noise and duplicates.

What's in the Dataset?

Basic **background** of our **data** (Main Page)

American Dialect Prediction Quiz

Take this quiz to discover which American dialect region you're from based on your word choices! Answer 10 questions about everyday terms and we'll predict your regional dialect.

Questions

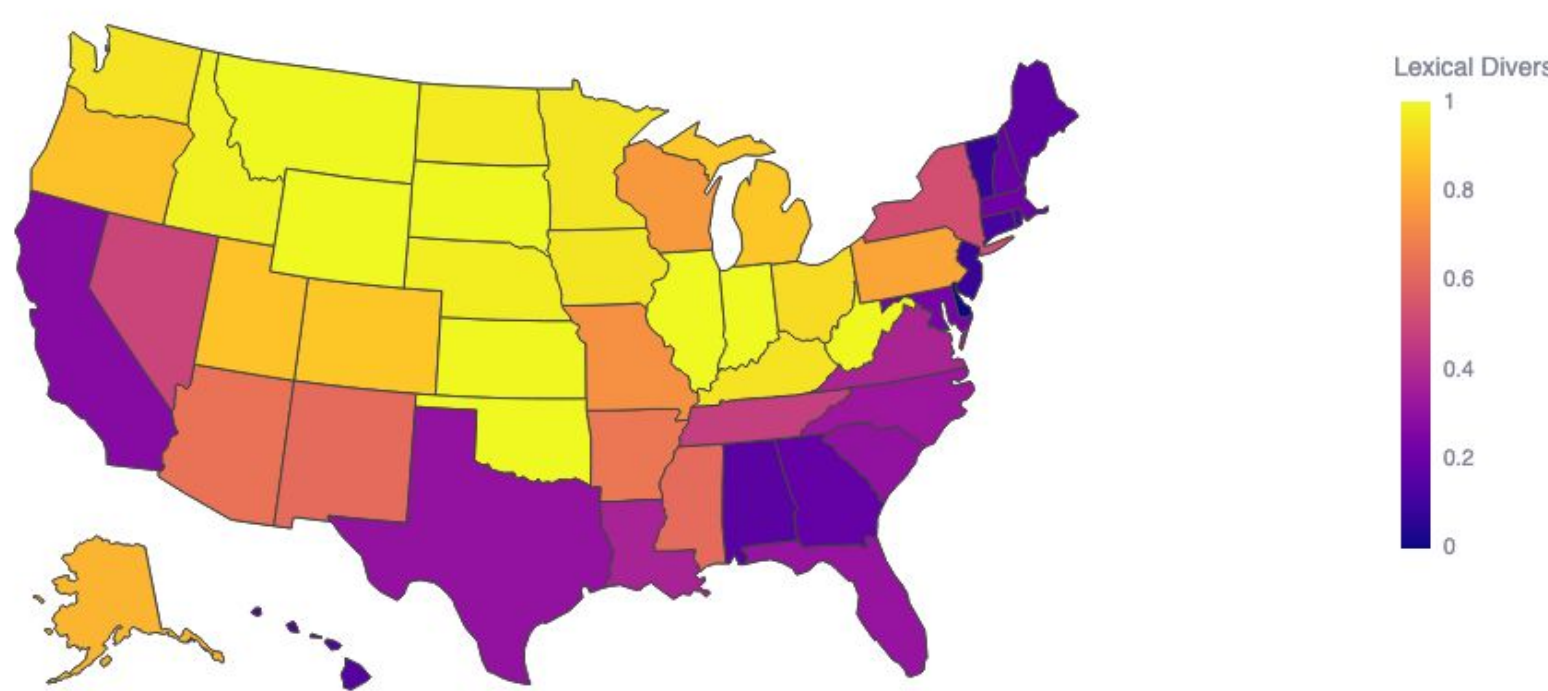
Question 1 of 10

What do you call the drink made with milk + ice cream?

Select your answer:

- ☐ milkshake/shake
- ☐ frappe
- ☐ cabinet
- ☐ velvet
- ☐ thick shake
- ☐ other

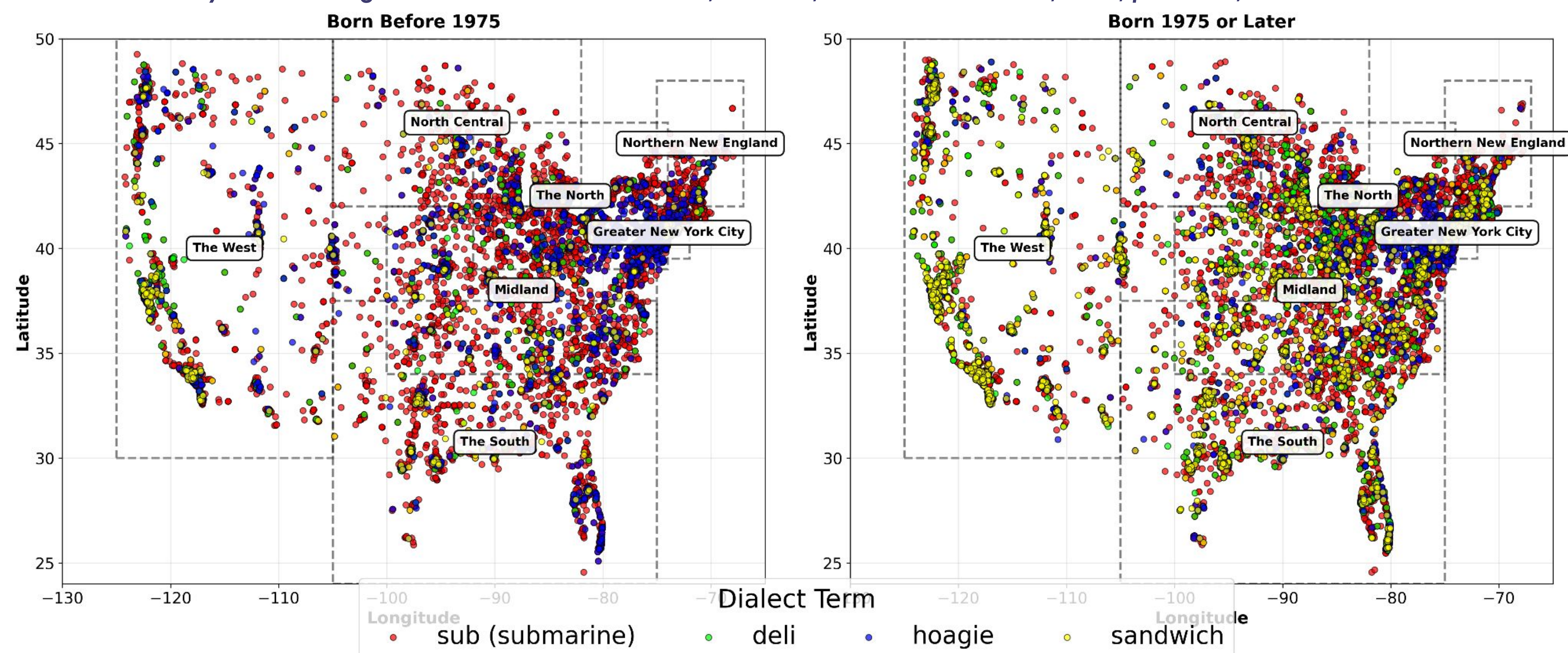
Dialect Prediction Quiz users can take (Prediction Page)



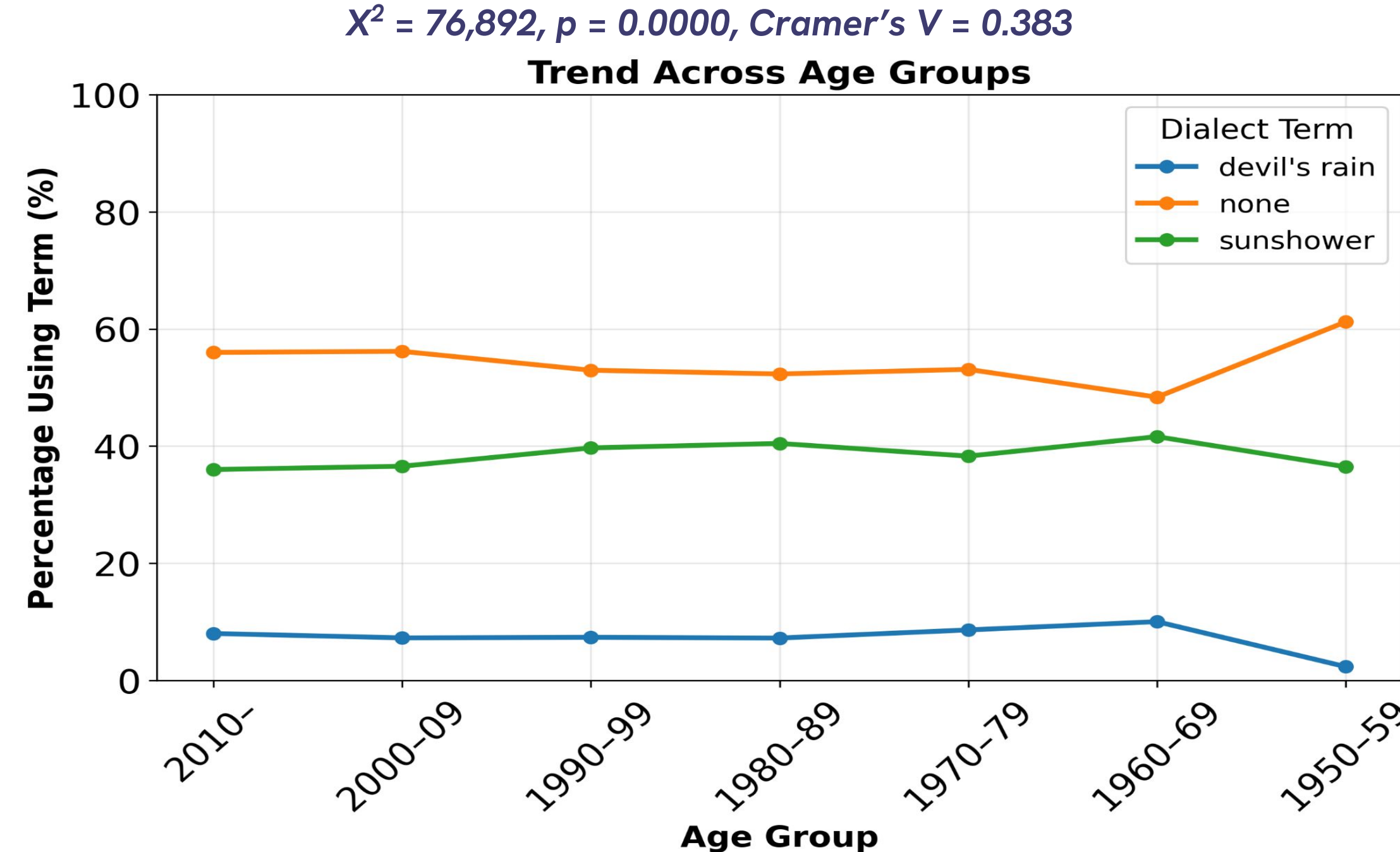
Example of a **Lexical Diversity** by U.S. State graph from our dashboard (Visualization Page)

Analysis: Change in Dialect Through Time/Region

Q: What do you call a long cold sandwich with cold cuts, lettuces, and so on? $X^2 = 126,545.3$, $p=0.0000$, Cramer's $V = 0.403$



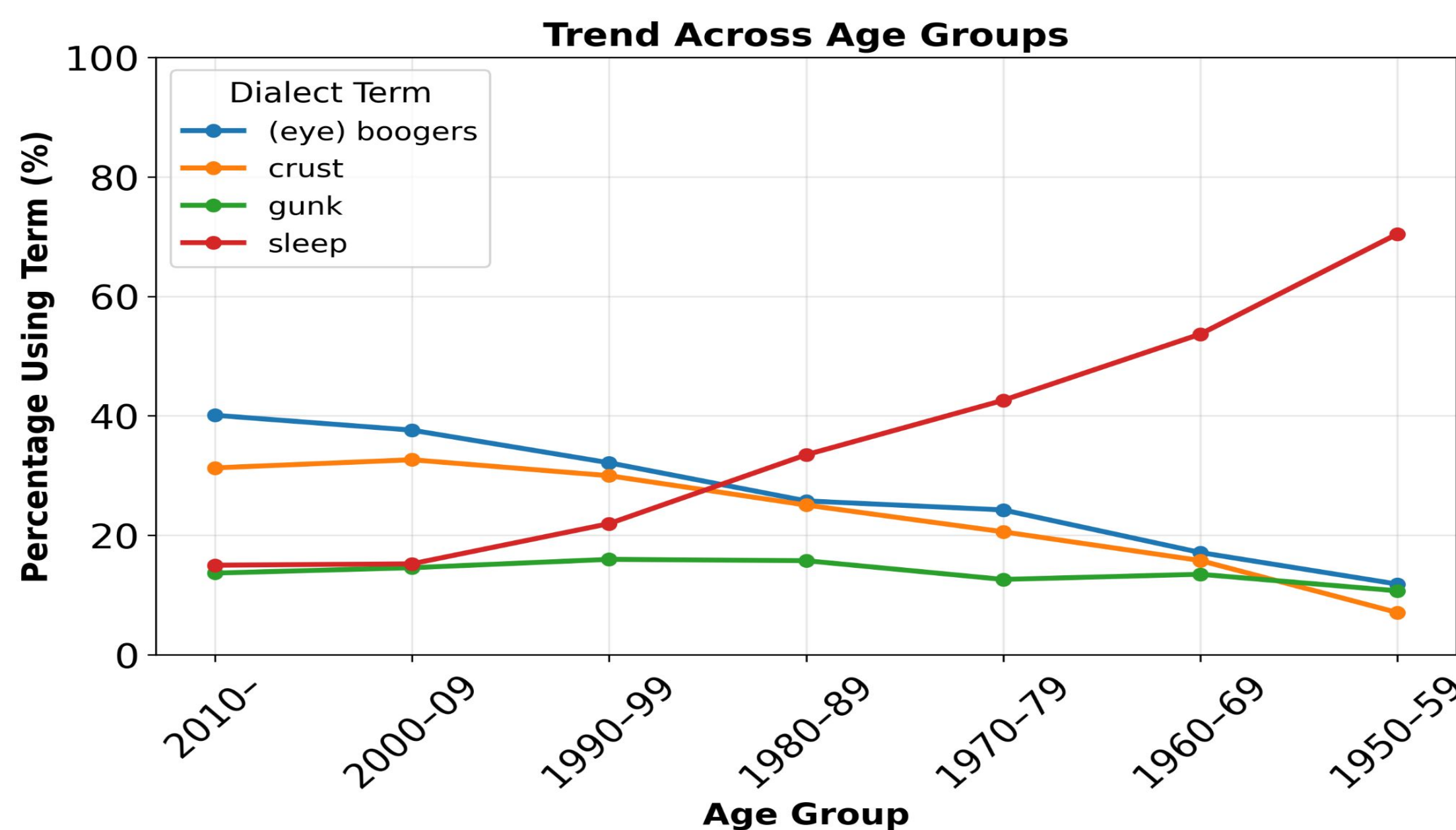
Q: What do you call the kind of rain when the sun is shining? $X^2 = 76,892$, $p = 0.0000$, Cramer's $V = 0.383$



Q: What do you call a small creature that rolls into a ball when you touch it? $X^2 = 79,231.3$, $p=0.0000$, Cramer's $V = 0.420$



Q: What do you call gooey or dry matter that collects in the corners of your eyes? $X^2 = 134,222.8$, $p = 0.0000$, Cramer's $V = 0.369$



- X^2 : Pattern or random → High (**pattern exists**)
- P-value**: Confidence in existence → Low (**high confidence**)
- Cramer's V**: Relationship strength of variables → Medium (**moderate**)

Conclusion & Next Steps



- Strong regional **patterns persist** for everyday words.



- Generational differences were **larger** than expected



- Models predicted regions well even without geographic data



- Add **more features** (pronunciation, syntax) to increase accuracy.



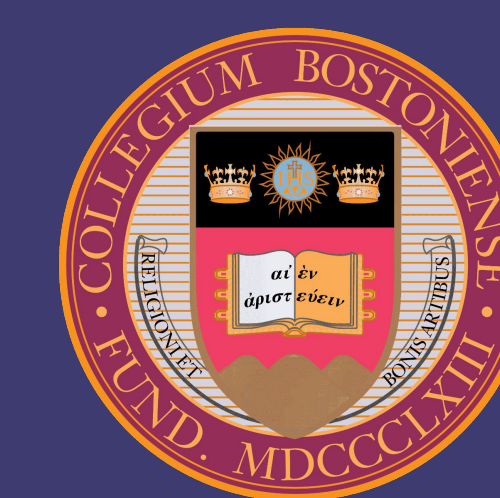
- Support **blended** or **multi-region** dialect predictions



- Improve handling of missing or inconsistent survey responses

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Data Processing

Ground-truth

Survey Question

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Answer Choices

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Respondent Answer

Sub (submarine)

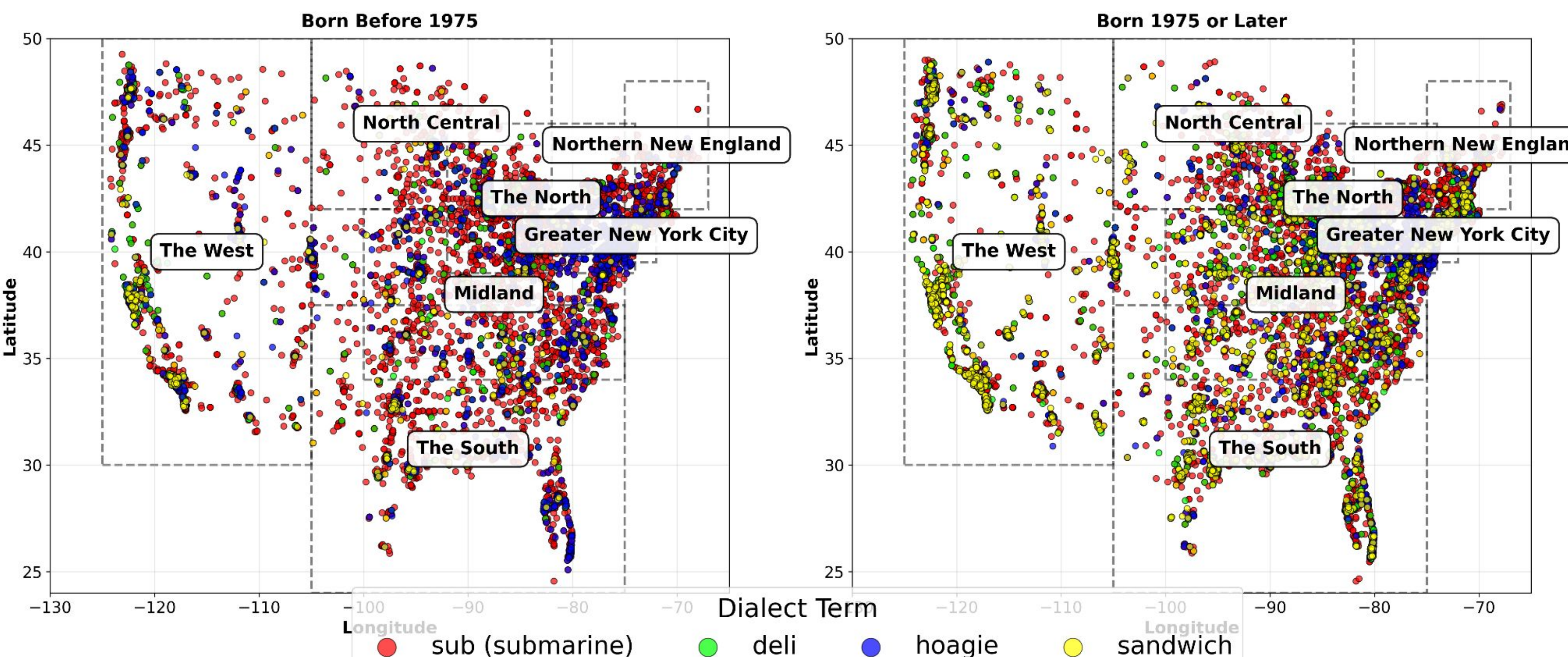
Example Data Point



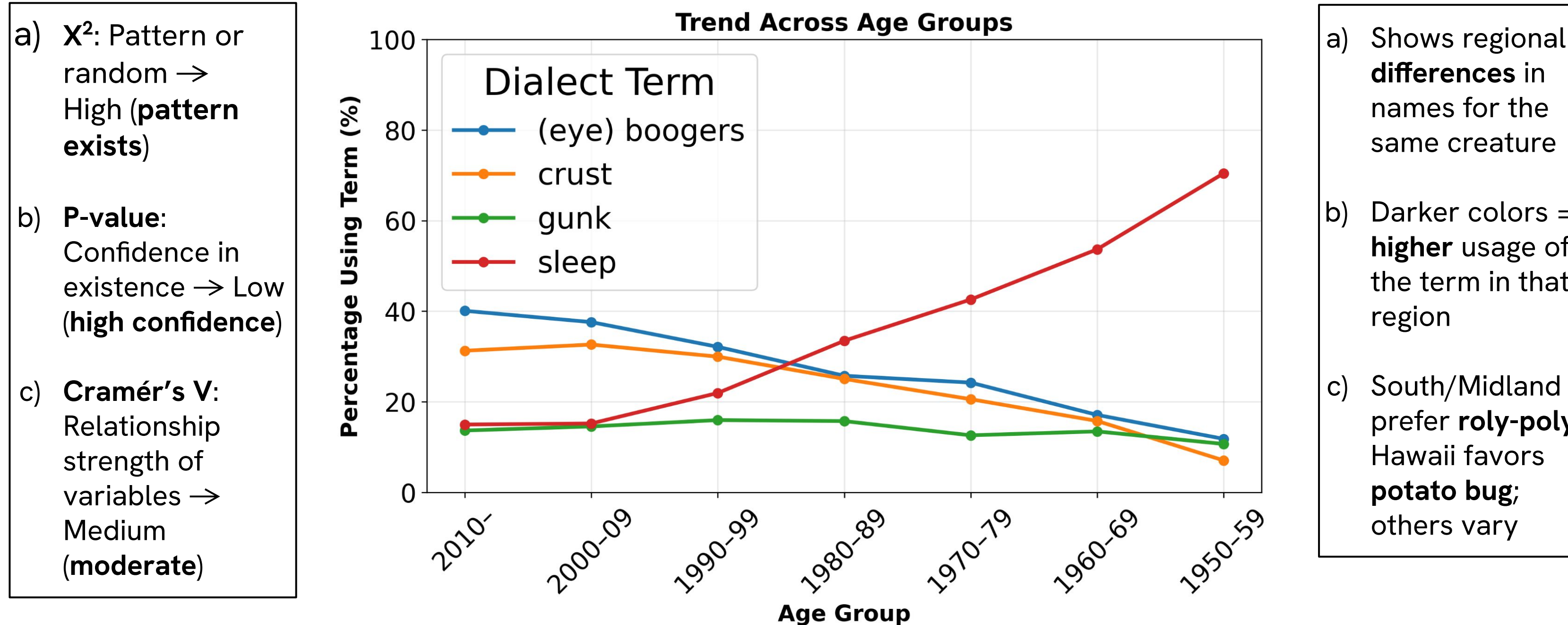
Labov, William, et al. The Atlas of North American English: Phonetics, Phonology and Sound Change: A Multimedia Reference Tool. Mouton de Gruyter, 2006.

Analysis: Change in Dialect Through Time/Region

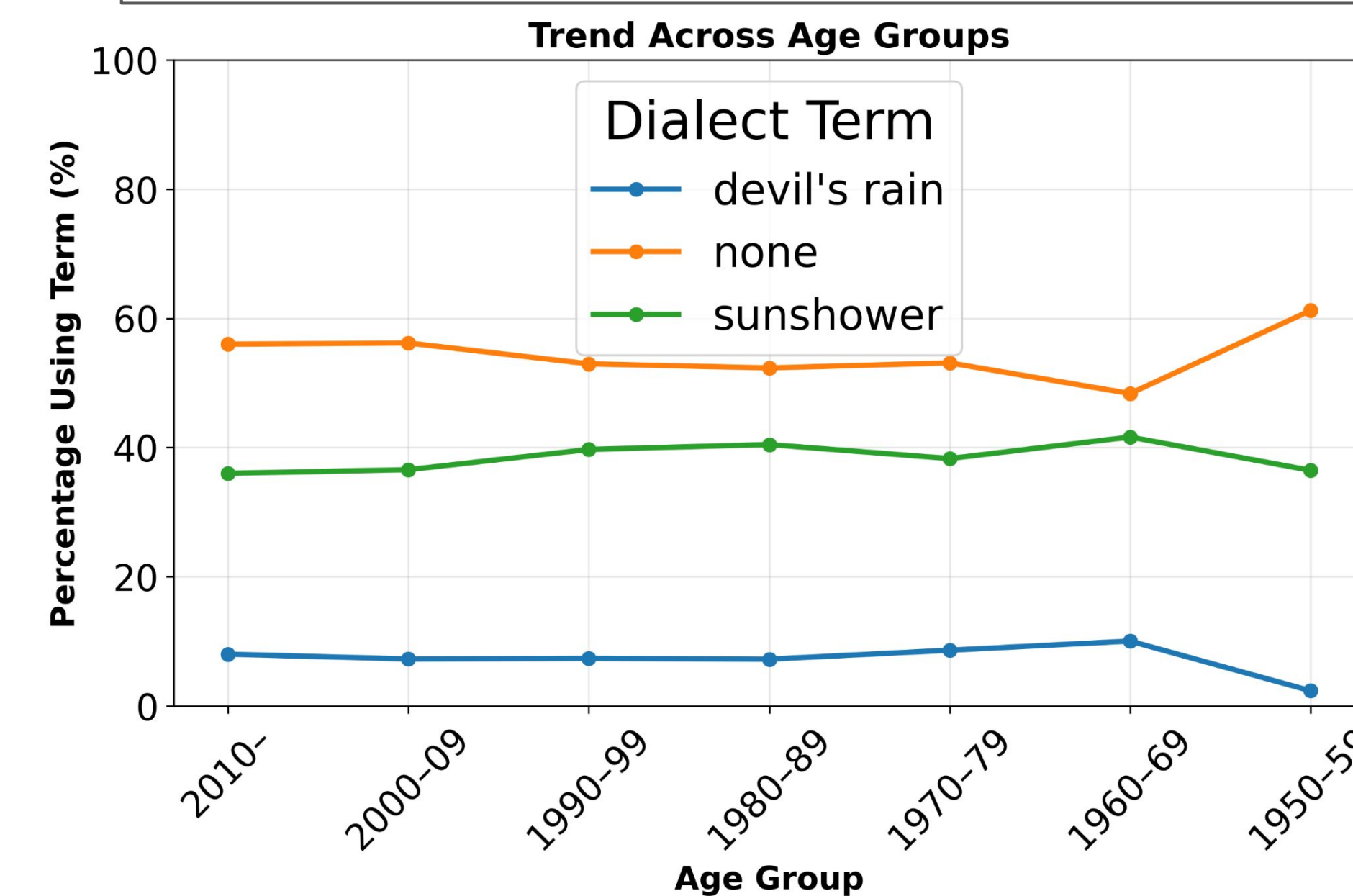
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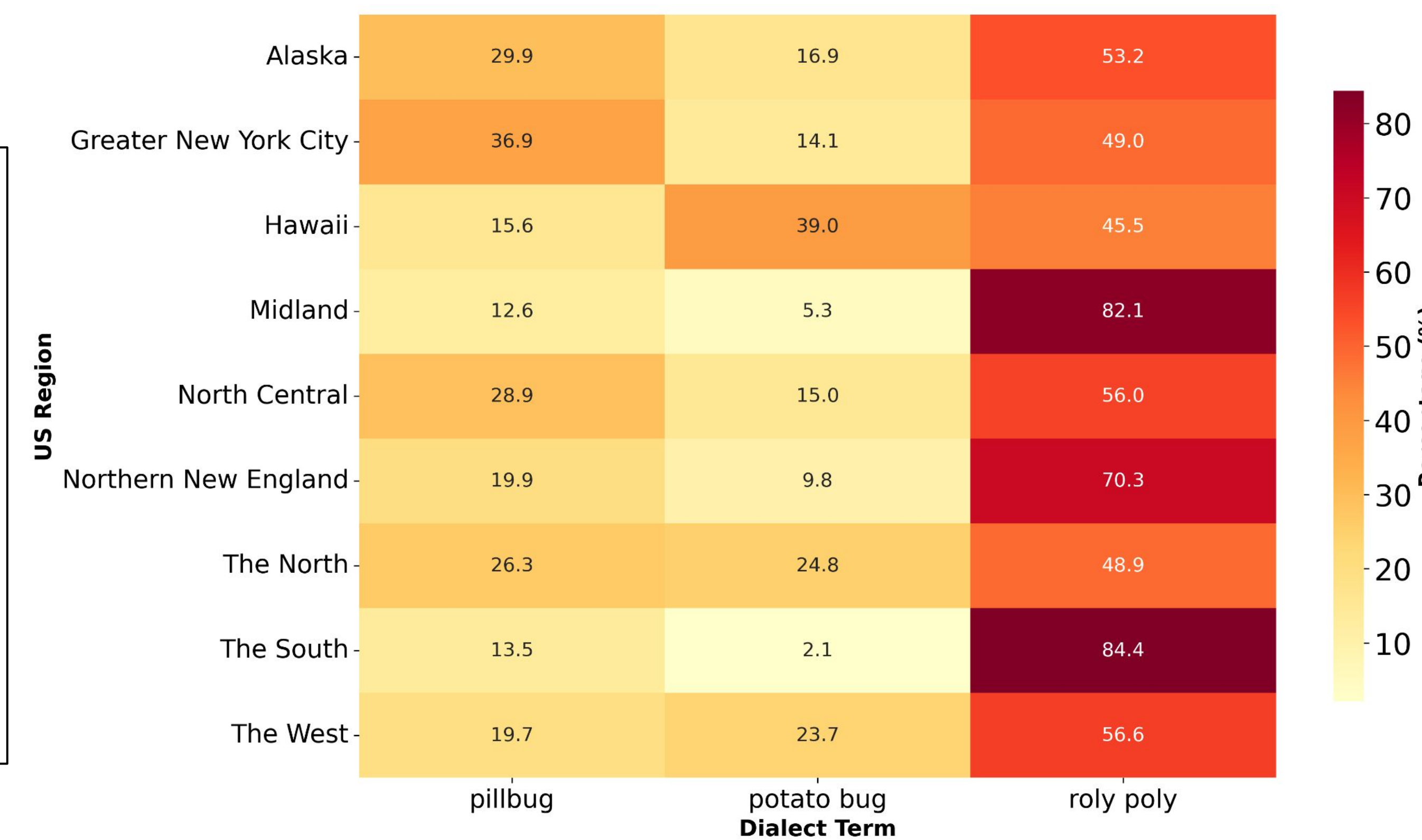
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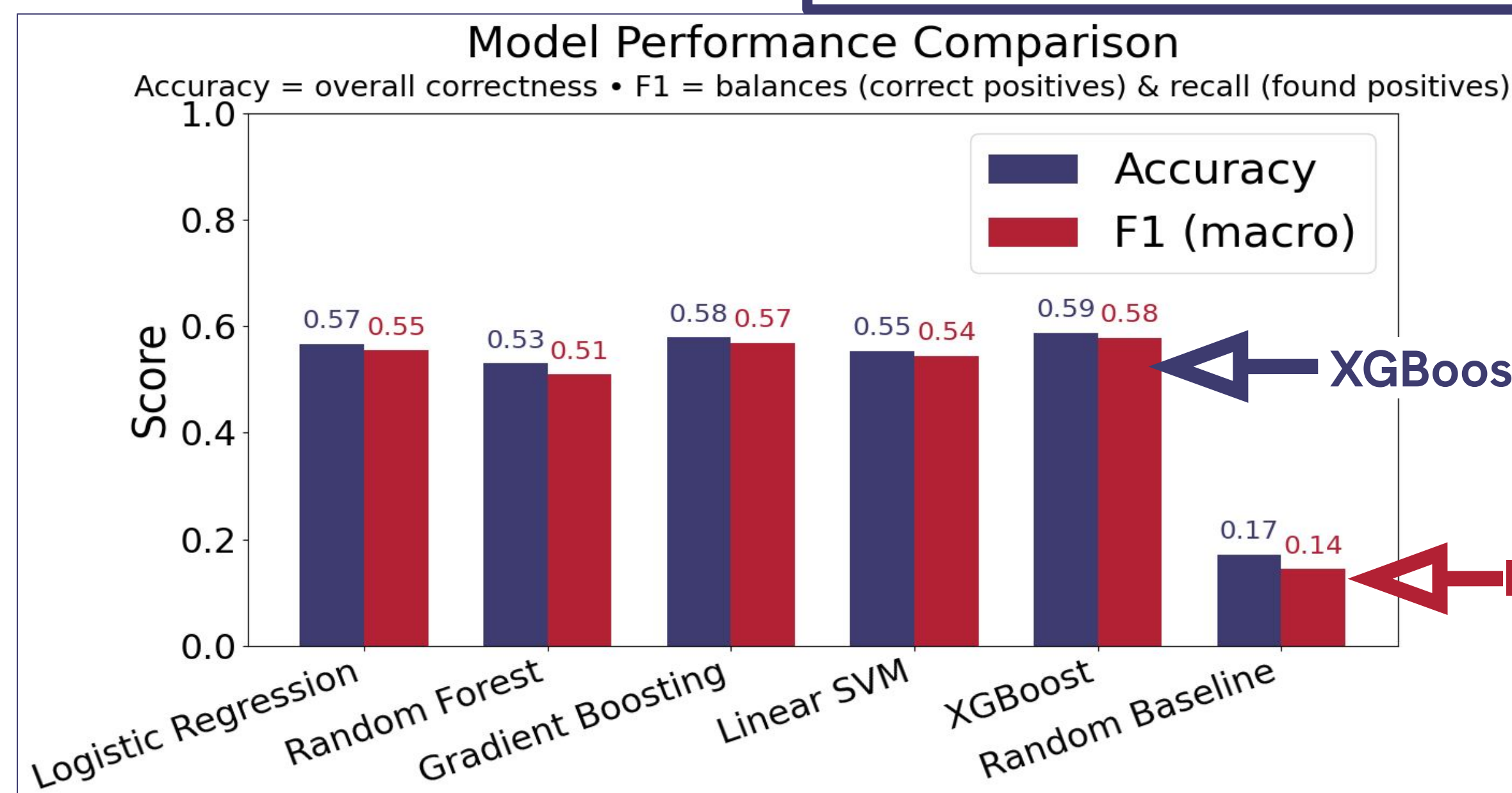
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Region Prediction Quiz (Dashboard)

Goal: Predict a user's region of origin based on their word choice

- One-hot-encoded question/answer pairs
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- Target variable: dialect region labels



XGBoost = Strongest

If we guessed.

Region	Strongest Predictor
Greater New York City	pronouncing Mary, merry, and marry all differently
Midland	calling the glowing summer insect a "firefly"
North Central	calling something diagonally across the street "kitty-corner"
Northern New England	pronouncing "route" to rhyme with "hoot"
The North	calling the glowing summer insect a "firefly"
The South	calling the school drinking fixture a "water fountain"
The West	using "highway" as the general term for a big, fast road

Dashboard (Live Demo 12/8)

Tech Stack

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- Interactive** visualizations
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American Dialect Prediction Quiz

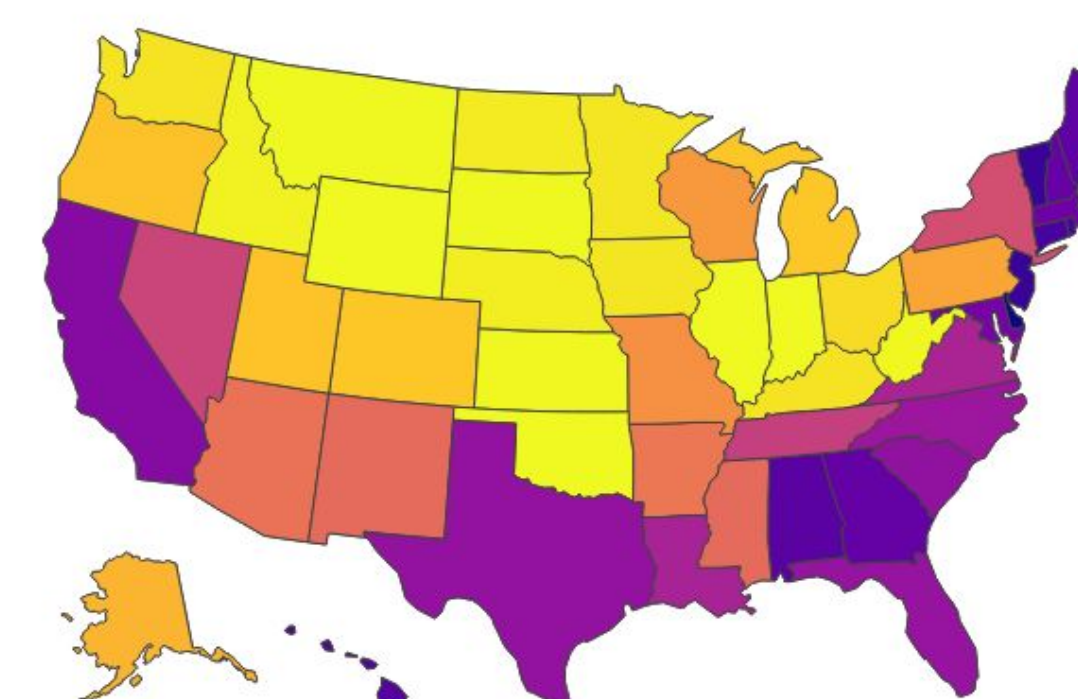
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Questions

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What do you call the drink made with milk + ice cream?

- Select your answer:
- ☐ milkshake/shake
 - ☐ frapp
 - ☐ cabernet
 - ☐ velvet
 - ☐ thick shake
 - ☐ other



Lexical Diversity

Example of a Lexical Diversity by each U.S. State graph from our dashboard

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